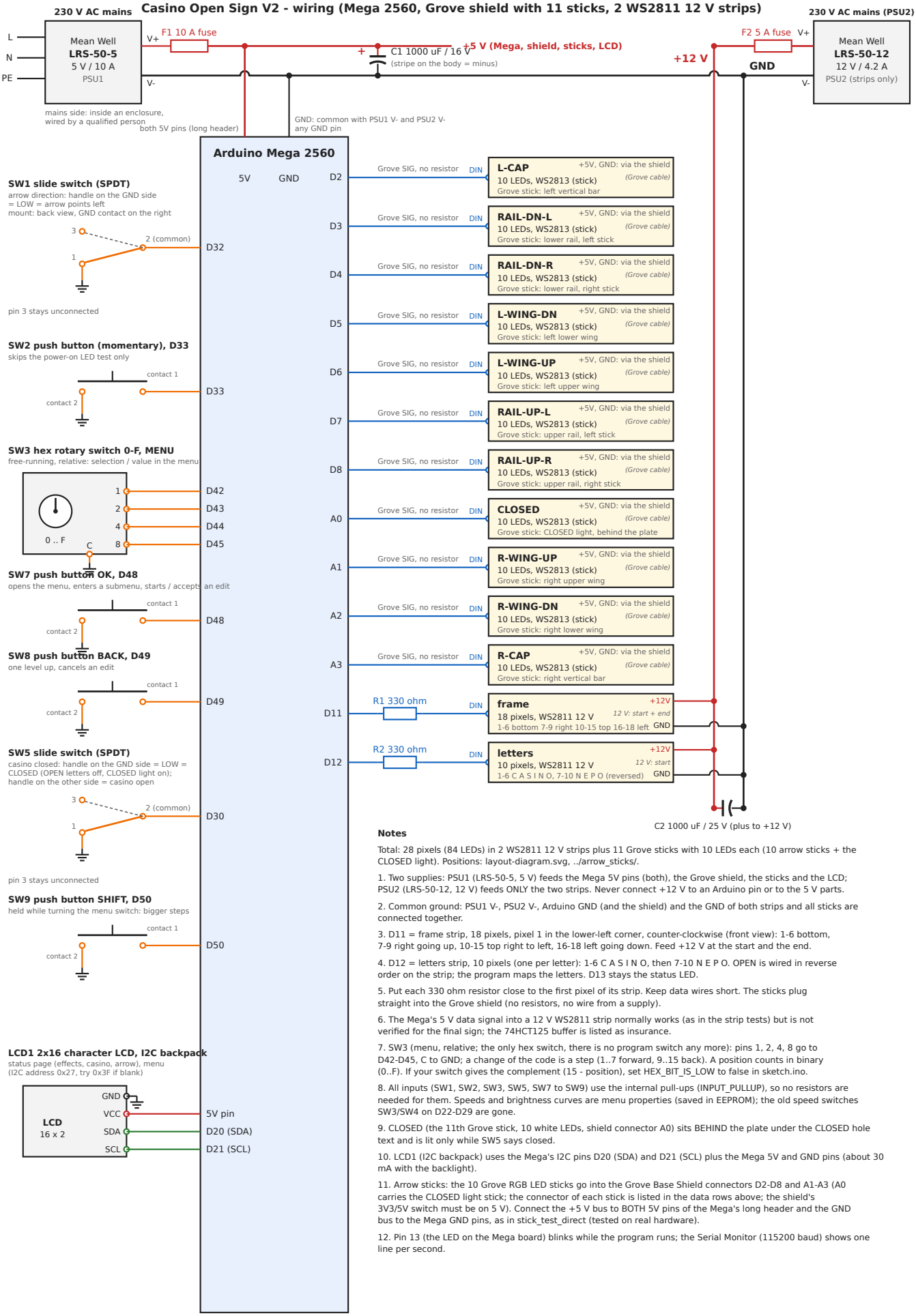
Casino Open Sign V2 - wiring (Mega 2560, Grove shield with 11 sticks, 2 WS2811 12 V strips)



Notes

Total: 28 pixels (84 LEDs) in 2 WS2811 12 V strips plus 11 Grove sticks with 10 LEDs each (10 arrow sticks + the CLOSED light). Positions: layout-diagram.svg, ../arrow_sticks/.

1. Two supplies: PSU1 (LRS-50-5, 5 V) feeds the Mega 5V pins (both), the Grove shield, the sticks and the LCD; PSU2 (LRS-50-12, 12 V) feeds ONLY the two strips. Never connect +12 V to an Arduino pin or to the 5 V parts.

2. Common ground: PSU1 V-, PSU2 V-, Arduino GND (and the shield) and the GND of both strips and all sticks are connected together.

3. D11 = frame strip, 18 pixels, pixel 1 in the lower-left corner, counter-clockwise (front view): 1-6 bottom, 7-9 right going up, 10-15 top right to left, 16-18 left going down. Feed +12 V at the start and the end.

4. D12 = letters strip, 10 pixels (one per letter): 1-6 C A S I N O, then 7-10 N E P O. OPEN is wired in reverse order on the strip; the program maps the letters. D13 stays the status LED.

5. Put each 330 ohm resistor close to the first pixel of its strip. Keep data wires short. The sticks plug straight into the Grove shield (no resistors, no wire from a supply).

6. The Mega's 5 V data signal into a 12 V WS2811 strip normally works (as in the strip tests) but is not verified for the final sign; the 74HCT125 buffer is listed as insurance.

7. SW3 (menu, relative; the only hex switch, there is no program switch any more): pins 1, 2, 4, 8 go to D42-D45, C to GND; a change of the code is a step (1..7 forward, 9..15 back). A position counts in binary (0..F). If your switch gives the complement (15 - position), set HEX_BIT_IS_LOW to false in sketch.ino.

8. All inputs (SW1, SW2, SW3, SW5, SW7 to SW9) use the internal pull-ups (INPUT_PULLUP), so no resistors are needed for them. Speeds and brightness curves are menu properties (saved in EEPROM); the old speed switches SW3/SW4 on D22-D29 are gone.

9. CLOSED (the 11th Grove stick, 10 white LEDs, shield connector A0) sits BEHIND the plate under the CLOSED hole text and is lit only while SW5 says closed.

10. LCD1 (I2C backpack) uses the Mega's I2C pins D20 (SDA) and D21 (SCL) plus the Mega 5V and GND pins (about 30 mA with the backlight).

11. Arrow sticks: the 10 Grove RGB LED sticks go into the Grove Base Shield connectors D2-D8 and A1-A3 (A0 carries the CLOSED light stick; the connector of each stick is listed in the data rows above; the shield's 3V3/5V switch must be on 5 V). Connect the +5 V bus to BOTH 5V pins of the Mega's long header and the GND bus to the Mega GND pins, as in stick_test_direct (tested on real hardware).

12. Pin 13 (the LED on the Mega board) blinks while the program runs; the Serial Monitor (115200 baud) shows one line per second.